THE BACK-AND-FORTH RELATION OF THE LINEAR ORDER TRANSFERS TO THE LABELLED TREE AT A SHIFT OF TWO

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Editorial note. The counted statement is the author environment at source0.tex lines 616-620 and its proof environment opens at line 621, the line immediately after; the pairing is by adjacency and the retained proof range 621-645 is the whole environment.

Editorial note. The retained preamble is the author style file dino.sty (package lines, the theorem declaration support, the notation and tuple macros and the theorem environments) followed by the ordinal and rank macros of the manuscript itself, reproduced verbatim; this is the same recipe as the previously published card of this article, so no package outside it is requested and extra_packages stays empty.

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Editorial note. No context span and no external reference are needed: the retained statement and proof use no label or citation from outside their own ranges.

Editorial scope. One original result of the pinned author manuscript, counted once: the transfer of the back-and-forth relation from the linear order of a structure to its labelled tree at a shift of two ranks (author label lem:bf_treetolo).

Allowed standard prerequisites (editorial).

- The linear order attached to a structure. A structure A carries a linear order L_A of labelled blocks, and the labelled tree T_A of its finite tuples is coded inside it.. It is the pair of objects the lemma relates.
- The coding map f. The map f sends the elements of L_A to the corresponding nodes of the labelled tree, so that comparing f applied to a tuple compares the coded tuples.. It is the map through which the two relations are compared.
- Back-and-forth relations of both structures. The rank alpha back-and-forth relations are defined by the game of that rank, and a rank comparison is decided by the interval relations left after each move.. They are the two relations shown to correspond at shifted ranks.

Lemma 0.1. *For every α and ordered $\bar{a}, \bar{b} \in L_{\mathcal{A}}$ such that if a_i, or b_i are from a labeled block, then every element in this block is in $\bar{a}$, respectively $\bar{b}$, $\bar{a} \leq_{3+\alpha} \bar{b}$ if and only if $f(\bar{a}) \leq_{1+\alpha} f(\bar{b})$.*

Proof. The proof is by transfinite induction with the only interesting case being the base case. For this note that the label and predecessor relation of $\mathcal{T}_{\mathcal{A}}$ in $L_{\mathcal{A}}$

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are both Σ_3^{in} and Π_3^{in} definable in $L_{\mathcal{T}_\mathcal{A}}$. Let φ be a Σ_1^{in} formula true of $\bar{a}$, then this formula can be translated into a Σ_3^{in} formula true of $f(\bar{a})$ and thus $\bar{a} \leq_3 \bar{b}$ implies $f(\bar{a}) \leq_1 f(\bar{b})$.

On the other hand, assume that $f(\bar{a}) \leq_1 f(\bar{b})$ and without loss of generality that $\bar{a}$ and $\bar{b}$ contain elements from exactly two different blocks. Let us consider the back-and-forth game where the $\forall$-player plays elements $\bar{b}^1$ on their first turn. For every b_i^1 the $\exists$-player looks for elements a_i^1 such that $l(f(a_i^1)) = l(f(b_i^1))$, a_i^1 is in the same position in its block as b_i^1 and $\bar{a}\bar{a}^1$ and $\bar{b}\bar{b}^1$ are isomorphic in the tree-ordering. Note that they will find a suitable $\bar{a}^1$ because $f(\bar{a}) \leq_1 f(\bar{b})$. Now on their last turn the $\forall$-player plays a tuple $\bar{a}^2$ and the $\exists$-player has to respond playing $\bar{b}^2$ such that $\bar{a}\bar{a}^1\bar{a}^2 \leq_1 \bar{b}\bar{b}^1\bar{b}^2$. All they need to do is ensure that the intervals in $\bar{a}\bar{a}^1\bar{a}^2$ are at least as large as the intervals in $\bar{b}\bar{b}^1\bar{b}^2$. Between any two elements in different blocks in $L_\mathcal{A}$, there are blocks coding paths in $\mathcal{T}_\mathcal{A}$. Thus, between any two elements in different blocks there are blocks of arbitrary large size and thus the $\exists$-player can find elements satisfying this requirement. $\square$